

VIET-MAN LE

Project Assistant and PhD Candidate

Institute of Software Engineering and Artificial Intelligence

Graz University of Technology, Austria

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RESEARCH INTERESTS

Knowledge-based Diagnosis, Configuration Systems, Software Product Lines, Explanations in AI, Large Language Models for Software Product Line Engineering, Constraint Acquisition.

EDUCATION

Graz University of Technology

Graz, Austria

PhD in Computer Science (in progress)

2020 – expected 2026

- Thesis: *Intelligent Techniques for Efficient Diagnostic Reasoning in Constraint-Based Systems*. Develops solver-agnostic algorithms for conflict detection, diagnosis, and quality assurance that scale to industrial constraint-based systems, such as configuration systems and recommender systems.
- Supervisor: Prof. Alexander Felfernig

Institute for Informatics in French-speaking areas

Hanoi, Vietnam

Master of Computer Science – Artificial Intelligence & Multimedia

2009 – 2011

- Double diplomas with University of La Rochelle (France) and Hanoi University of Science and Technology (Vietnam)
- Master's Thesis: *Système de navigation dans des bases de données d'images* (image browser based on Self-Organizing Maps) – Score: 15.4/20 (Très bien). Supervisor: Prof. Michel Verleysen, Catholic University of Louvain, Louvain-la-Neuve, Belgium
- Master 1 TPE: *Visualisation de graphes pour la recherche interactive d'images* (graph-based visualization for interactive image retrieval) – Score: 15/20 (ranked 1st in class). Supervisor: Prof. Alain Boucher

University of Sciences, Hue University

Hue, Vietnam

Bachelor of Computer Science

2000 – 2004

- Bachelor's Thesis: *Check Spelling Based on Analyzing the Grammar of Vietnamese Phrases* – Score: 10/10 (one of three students with maximum thesis score). Supervisor: Prof. Nguyen Gia Dinh
- Graduated in the top 10 of 150 students

EMPLOYMENT

Graz University of Technology

Graz, Austria

University Assistant/Project Assistant and PhD Candidate

2020 – present

- Designed and implemented solver-agnostic algorithms for conflict detection, direct diagnosis, redundancy detection, and automated debugging that scale to industrial constraint-based systems, such as configuration systems and recommender systems
- Developed and maintained open-source research software (HiConfiT, DirectDebug, FM2ExConf) supporting feature model analysis and diagnosis, with contributions integrated into the flamapy ecosystem
- Co-supervised 5 bachelor and master theses, two of which yielded peer-reviewed publications (AAAI 2026, PAIS/ECAI 2023)

University of Economics, Hue University

Hue, Vietnam

Vice-Dean, Faculty of Economic Information Systems

2014 – 2019

- Led the design and launch of the Undergraduate Degree Program in Management Information Systems
- Established faculty-level policies for academic advising, career planning, and student professionalisation
- Coordinated curriculum development, teaching assignments, and quality assurance across the faculty's undergraduate programmes

University of Economics, Hue University

Hue, Vietnam

Lecturer, Faculty of Economic Information Systems

2004 – 2020

- Taught Programming (C#, Python, Java), Data Structures and Algorithms, and Windows Applications Programming to undergraduate students
- Researched recommender systems, with a focus on multi-agent models for collaborative filtering
- Advised 2–6 undergraduate students per year (2012–2019) on bachelor theses and student research projects in information systems

FUNDED RESEARCH PROJECTS

- GenRE – Generative AI for Requirements Engineering** 2024 – 2027
FFG Bridge project. Team member. Industry partner: Morgendigital.
- Develops LLM-based techniques for requirements elicitation, quality assurance, and validation in software engineering
- Kickscale: KI-Engine – Sales-Specialized LLM for the Kickscale SaaS Platform** 2023 – 2025
FFG Basisprogramm project. Scientific consulting role via TU Graz. Industry lead: Kickscale GmbH.
- Developed LLM-based methods for extracting Ideal Customer Profiles from CRM deal data, and released frameworks for evaluating LLM and RAG outputs
- OpenSpace – AI Techniques for Testing Highly-Variant Software** 2021 – 2024
FFG Bridge project. Team member. Industry partner: Uniquare GmbH.
- Developed machine-learning approaches for testing and debugging variability-intensive software, including automated analysis of variability models and identification of faulty components
- ParXCel – Scalable Constraint Solving via ML and Parallelization** 2020 – 2023
FFG Bridge project. Team member. Industry partner: Combeeneration GmbH.
- Integrated machine learning into constraint-based reasoning and parallelized conflict detection and direct diagnosis to boost performance (FastDiagP, InformedQX)

SELECTED PUBLICATIONS

According to [Google Scholar](#) (May 2026): **775 citations**, h-index **13**, i10-index **17**, across more than 50 peer-reviewed papers. Full list at manleviet.github.io/publications.html.

- Christian Bähnisch, Alexander Felfernig, Damian Garber, Albert Haag, Denis Helic, **Viet-Man Le**, Sebastian Lubos. *Machine Learning for Constraint-based Configuration: A Survey*. *Journal of Artificial Intelligence Research* (to appear), 2026. **SCIMAGO Q1**
- Viet-Man Le**, Lukas A. Feldgrill, Alexander Felfernig. *Robust Lazy Conflict Detection via Multi-Conflict Extraction and Genetic Diversity Control*. *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 23, pp. 19208–19215, 2026. **CORE A***
- Viet-Man Le**, Thi Ngoc Trang Tran, Sebastian Lubos, Alexander Felfernig, Damian Garber. *Early-Stage Product Line Validation Using LLMs: A Study on Semi-Formal Blueprint Analysis*. *Proceedings of the 41st ACM/SIGAPP Symposium on Applied Computing (SAC 2026)*, pp. 1620–1627, 2026. **CORE B**
- Mathias Uta, **Viet-Man Le**, Alexander Felfernig, Denis Helic. *Learning Constraint Orderings for Direct Diagnosis*. *Journal of Intelligent Information Systems*, vol. 63, no. 5, pp. 1753–1777, 2025. **SCIMAGO Q2**
- Viet-Man Le**, Alexander Felfernig, Thi Ngoc Trang Tran, Mathias Uta. *InformedQX: Informed Conflict Detection for Over-Constrained Problems*. *Proceedings of the 38th AAAI Conference on Artificial Intelligence*, vol. 38, no. 9, pp. 10616–10623, 2024. **CORE A***
- Viet-Man Le**, Cristian Vidal Silva, Alexander Felfernig, David Benavides, José Galindo, Thi Ngoc Trang Tran. *FastDiagP: An Algorithm for Parallelized Direct Diagnosis*. *Proceedings of the 37th AAAI Conference on Artificial Intelligence*, vol. 37, no. 5, pp. 6442–6449, 2023. **CORE A***
- Viet-Man Le**, Alexander Felfernig, Mathias Uta, David Benavides, José Galindo, Thi Ngoc Trang Tran. *DirectDebug: Automated Testing and Debugging of Feature Models*. *Proceedings of the ACM/IEEE 43rd International Conference on Software Engineering: New Ideas and Emerging Results*, pp. 81–85, 2021. **CORE A***

GRANTS & AWARDS

- Projektfonds Lehre 2026 (Project Fund for Teaching)** 2026
TU Graz Vice-Rectorate for Teaching – MIRROR project (team member, PI: Prof. Alexander Felfernig)
- Best Teaching Award – Nominations** 2023, 2024, 2025
Course Introduction to Structured Programming, Graz University of Technology
- Young Experts: Minute Madness Session** 2024
Austrian Computer Science Day 2024 – session for excellent doctoral students in CS at Austrian universities
- Hue University Research Grant** 2015 – 2016
Project: Multi-agent model for tourist attraction recommender systems (team member)
- Internship Grant, Wallonie-Bruxelles International** 2011
Supporting a 7-month research internship at Catholic University of Louvain, Belgium
- AUF Master’s Scholarship** 2008 – 2011
Agence Universitaire de la Francophonie – Master of Computer Science program at IFI, Vietnam
- Outstanding Student Award in Computer Science** 2004
Ministry of Science and Technology of Vietnam – top 100 CS students nationwide, sponsored by Motorola

SERVICE

Conference and Workshop Organisation

Chair, 27th International Workshop on Configuration (ConfWS 2025) <i>Co-located with the 28th European Conference on Artificial Intelligence (ECAI 2025), Bologna, Italy</i>	2025
Proceedings Chair, 26th ACM International Systems and Software Product Line Conference <i>Graz, Austria</i>	2022

Program Committee Member

2025: RecSys 2025, SPLC 2025, ECAI 2025, AAAI 2026
2024: ConfWS 2024, IntRS 2024, SPLC 2024

Conference and Journal Reviewer

2026: 1 × SoSyM, 1 × JIIS
2025: 2 × RecSys 2025, 2 × SPLC 2025, 4 × ECAI 2025, 5 × AAAI 2026, 7 × JIIS
2024: 2 × ConfWS 2024, 2 × IntRS 2024, 3 × SPLC 2024, 2 × ESWA

TEACHING

Graz University of Technology (University Assistant)

Introduction to Structured Programming (ISP) <i>Course organization, assignment design. Nominated for Best Teaching Award in 2023, 2024, 2025.</i>	2020 – 2026
Object-Oriented Analysis and Design (OAD) <i>Assignment design on design principles and patterns; lecturer on Design Patterns in 2024 and 2025.</i>	2022 – 2026

University of Economics, Hue University (Lecturer, 2004 – 2019)

Courses taught: Introduction to Programming (C#, Python), Windows Applications Programming, Data Structures and Algorithms, Object-Oriented Programming.

MENTORING AND CO-SUPERVISION

Fabian True (Bachelor) <i>FM2ExConf: A Web-Service to Convert Feature Models into an Executable Representation in Microsoft Excel</i>	2025
Nhat Minh Hoang (Bachelor) <i>From Conflicts to Efficiency: Learning Constraint Orderings for QuickXPlain</i>	2025
Lukas A. Feldgrill (Master) <i>Robust Lazy Conflict Detection via Multi-Conflict Extraction and Genetic Diversity Control (AAAI 2026)</i>	2025
Yunus Emre Akca (Bachelor) <i>PWipeOutR: Parallelized Redundancy Detection for Feature Models</i>	2023
Tamim Burgstaller (Bachelor) <i>FMTesting: A FeatureIDE Plug-in for Automated Feature Model Analysis and Diagnosis (PAIS/ECAI 2023)</i>	2022

SOFTWARE & TOOLS

HiConfiT – Suite of open-source Java libraries (*hiconfit-core*) and CLI apps (*KBStatistics*, *FMGen*) for Knowledge-Based Configuration Systems.
FMTesting – FeatureIDE plug-in for automated feature model testing and debugging (PAIS 2023).
FM2ExConf – Feature Model to Excel Configurator converter, built on hiconfit-core.
DirectDebug – Java software package for automated testing and debugging of variability models (Software Impacts, 2021).
flamapy – Contributed FastDiagP, DirectDebug, and WipeOutR plugins to the Python ecosystem for feature model analysis.

TECHNICAL SKILLS

Constraint Solvers & Tools: Choco, SAT, MiniZinc
Programming Languages: Java, Python, C/C++, SQL
Frameworks & DevOps: Spring Boot, FeatureIDE (Eclipse plug-in), Git, Docker, GitHub Actions
Languages: English (fluent), Vietnamese (native)

REFERENCES

Univ.-Prof. Dipl.-Ing. Dr.techn. Alexander Felfernig

PhD Supervisor

Full Professor, Institute of Software Engineering and AI, Graz University of Technology

- Address: Inffeldgasse 16b/II, 8010 Graz, Austria
- Email: alexander.felfernig@tugraz.at

Assoc. Prof. Dr. Truong Tan Quan

Long-term Colleague

Rector, University of Economics, Hue University, Vietnam

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Dr. Habil. Binh Nguyen

Former Undergrad. Lecturer

Research Scholar, International Institute for Applied Systems Analysis (IIASA), Austria

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